

Practical Work Gathering Information about Linux and Wget

Purpose

- To learn basic information gathering about Linux systems
- To practice downloading files using the `wget` utility
- To check packages for integrity

1. Basic Information Gathering

Distribution Information

```
cat /etc/*release
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$ cat /etc/*release
DISTRIB_ID=Ubuntu
DISTRIB_RELEASE=22.04
DISTRIB_CODENAME=jammy
DISTRIB_DESCRIPTION="Ubuntu 22.04.5 LTS"
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$
```

Kernel Information

```
uname -a
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$ uname -a
Linux DESKTOP-6966M0H 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon Dec 1 20:46:23 UTC 2025 x86_64 x86_64
4 x86_64 GNU/Linux
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$
```

Current User, Uptime, Directory Permissions

Whoami, uptime, `ls -la /home`

```
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$ whoami
christian
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$ uptime
 22:13:05 up 5 min,  1 user,  load average: 0.00, 0.00, 0.00
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$ ls -la /home
total 12
drwxr-xr-x  3 root      root      4096 Jun  1 15:32 .
drwxr-xr-x 24 root      root      4096 Jun  3 22:08 ..
drwxr-x---  3 christian  christian 4096 Jun  1 15:59 christian
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$
```

```
sudo chmod o-rx /home/kali
```

```
christian@DESKTOP-6966M0H: /mnt/c/Users/T301-Sec-198/Desktop$ sudo chmod o-rx /home/ubuntu
[sudo] password for christian:
Sorry, try again.
[sudo] password for christian:
chmod: cannot access '/home/ubuntu': No such file or directory
christian@DESKTOP-6966M0H: /mnt/c/Users/T301-Sec-198/Desktop$ sudo chmod o-rx /home/ubuntu
```

Network Neighbors

```
ip neigh
```

```
christian@DESKTOP-6966M0H: /mnt/c/Users/T301-Sec-198/Desktop$ ip neigh
.176.1 dev eth0 lladdr 08:00:27:43:45:fb REACHABLE
christian@DESKTOP-6966M0H: /mnt/c/Users/T301-Sec-198/Desktop$
```

Listening Services

```
sudo ss -tlnp
```

```
christian@DESKTOP-6966M0H: /mnt/c/Users/T301-Sec-198/Desktop$ sudo ss -tlnp
State      Recv-Q      Send-Q      Local Address:Port
LISTEN     0            1000        0.0.0.0:255.254:53
LISTEN     0            4096        0.0.0.0:0.53%lo:53
resolve",pid=132,fd=14))
christian@DESKTOP-6966M0H: /mnt/c/Users/T301-Sec-198/Desktop$
```

Shows active services and ports (e.g., sshd, apache2, vsftpd).

Sudo Privileges

```
sudo -l
```

```
christian@DESKTOP-6966M0H: /mnt/c/Users/T301-Sec-198/Desktop$ sudo -l
Matching Defaults entries for christian on DESKTOP-6966M0H:
  env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

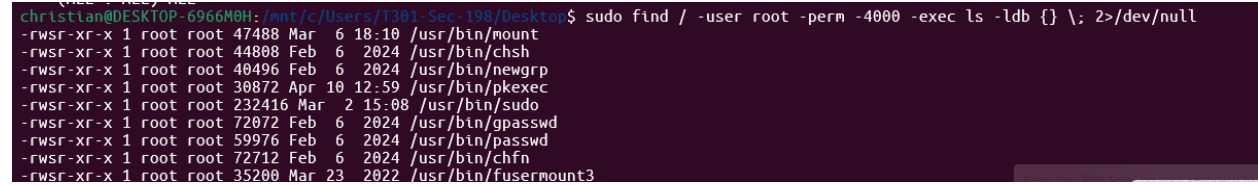
User christian may run the following commands on DESKTOP-6966M0H:
  (ALL : ALL) ALL
christian@DESKTOP-6966M0H: /mnt/c/Users/T301-Sec-198/Desktop$
```

Lists commands the user can run with sudo. Example: (ALL : ALL) ALL.

2. SUID Files

SUID/SGID allow executables to run with elevated privileges. Find them with:

```
sudo find / -user root -perm -4000 -exec ls -ldb {} \; 2>/dev/null
```



The terminal screenshot shows the command being executed and its output. The output lists several files with permissions, owner, group, size, date, and path. The files are: /usr/bin/mount, /usr/bin/chsh, /usr/bin/newgrp, /usr/bin/pkexec, /usr/bin/sudo, /usr/bin/gpasswd, /usr/bin/passwd, /usr/bin/chfn, and /usr/bin/fusermount3.

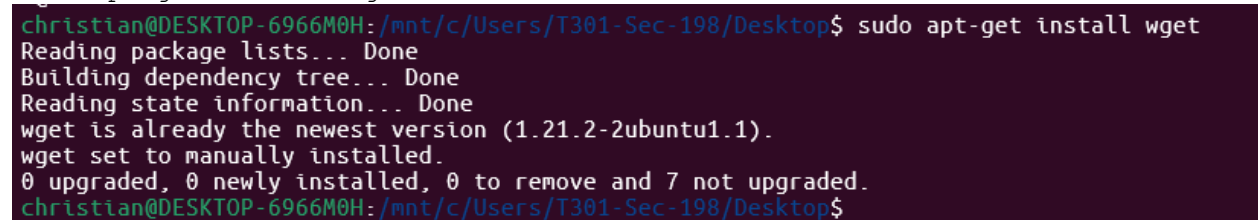
Examples include:

- -rwsr-xr-x 1 root root 47488 Mar 6 18:10 /usr/bin/mount
- -rwsr-xr-x 1 root root 44808 Feb 6 2024 /usr/bin/chsh
- -rwsr-xr-x 1 root root 40496 Feb 6 2024 /usr/bin/newgrp
- -rwsr-xr-x 1 root root 30872 Apr 10 12:59 /usr/bin/pkexec
- -rwsr-xr-x 1 root root 232416 Mar 2 15:08 /usr/bin/sudo
- -rwsr-xr-- 1 root messagebus 35112 Oct 25 2022 /usr/lib/dbus-1.0/dbus-daemon-launch-helper
- -rwsr-xr-- 1 root landscape 14488 Jun 13 2025 /usr/lib/landscape/apt-update
- -rwsr-xr-x 1 root root 338536 Apr 28 02:38 /usr/lib/openssh/ssh-keysign
- -rwsr-xr-x 1 root root 22832 Apr 10 12:59 /usr/libexec/polkit-agent-helper-1

3. Downloading and Using Wget

Install wget

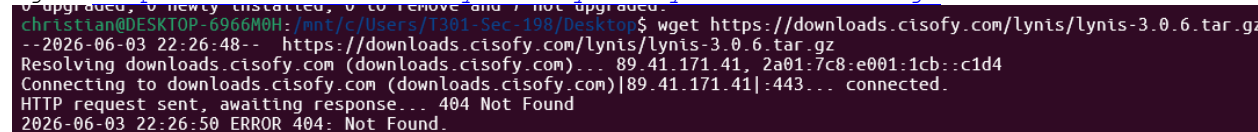
```
sudo apt-get install wget
```



The terminal screenshot shows the command being executed and its output. The output indicates that wget is already the newest version (1.21.2-2ubuntu1.1) and is set to manually installed. It also shows that 0 packages were upgraded, 0 were newly installed, 0 were to be removed, and 7 were not upgraded.

Download Lynis

```
wget https://downloads.cisofy.com/lynis/lynis-3.0.6.tar.gz
```



The terminal screenshot shows the command being executed and its output. The output indicates that the file was successfully downloaded. It also shows the file size and the date and time of the download.

Not found but the command is correct

Extract Archive

unzip files.zip

```
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$ ls
'Antigravity IDE.lnk'      api_checker.py
'Antigravity.lnk'         desktop.ini
'Docker Desktop.lnk'      files
'Infrastructure security compliance' files.zip
'Microsoft Edge.lnk'      linux-info-gathering-wget
'Practical Work Gathering Information about Linux.docx' '~$actical Work Gathering Information
'Weekly Report'

christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$ unzip files.zip
Command 'unzip' not found, but can be installed with:
sudo apt install unzip
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$ sudo apt install unzip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Verify Integrity

sha256sum files.zip

```
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$ sha256sum files
files/      files.zip
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$ sha256sum files.zip
d4e6603d29ff589c57e9d8cae477f8a36b935bf434d9100f4387717f0d72910f  files.zip
christian@DESKTOP-6966M0H:/mnt/c/Users/T301-Sec-198/Desktop$
```

Expected hash: d4e6603d29ff589c57e9d8cae477f8a36b935bf434d9100f4387717f0d72910f

4. Package Integrity Checks

List Installed Packages

dpkg -l

```
root@DESKTOP-6966M0H:/# dpkg -l
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name                  Version                Architecture
+++-+-----+-----+-----+-----+
ii  adduser                  3.118ubuntu5           all
ii  adwaita-icon-theme       41.0-1ubuntu1          all
ii  apparmor                 3.0.4-2ubuntu2.5       amd64
ii  apport                  2.20.11-0ubuntu82.10   all
ii  apport-symptoms          0.24                   all
ii  apt                      2.4.14                 amd64
ii  apt-utils                2.4.14                 amd64
ii  at-spi2-core             2.44.0-3               amd64
ii  base-files               12ubuntu4.7            amd64
ii  base-passwd              3.5.52build1           amd64
ii  bash                     5.1-6ubuntu1.1         amd64
ii  bash-completion          1:2.11-5ubuntu1        all
ii  bc                       1.07.1-3build1         amd64
ii  bind9-dnsutils           1:9.18.39-0ubuntu0.22.04.4 amd64
ii  bind9-host               1:9.18.39-0ubuntu0.22.04.4 amd64
ii  bind9libs:amd64          1:9.18.39-0ubuntu0.22.04.4 amd64
ii  binutils                 2.38-4ubuntu2.12       amd64
ii  binutils-common:amd64    2.38-4ubuntu2.12       amd64
```

Show Files of a Package

```
dpkg -L coreutils
```

```
root@DESKTOP-6966M0H:/# dpkg -L coreutils
/
/bin
/bin/cat
/bin/chgrp
/bin/chmod
/bin/chown
/bin/cp
/bin/date
/bin/dd
/bin/df
/bin/dir
/bin/echo
/bin/false
/bin/ln
```

5. Incident Investigation

Find modified files:

```
sudo find / -printf "%T@;%Tc;%p\n"
```

```
root@DESKTOP-6966M0H:/# sudo find / -printf "%T@;%Tc;%p\n"
1780517291.3208597730;Wed Jun 3 22:08:11 2026;/
1780504918.0860470850;Wed Jun 3 18:41:58 2026;/wsl$Akigh
1773065771.6752449050;Mon Mar 9 16:16:11 2026;/bin
1773065773.0922711540;Mon Mar 9 16:16:13 2026;/opt
1780504918.0900470850;Wed Jun 3 18:41:58 2026;/wsl$DGKpjh
1773065771.6792449800;Mon Mar 9 16:16:11 2026;/lib
1773065773.1102714830;Mon Mar 9 16:16:13 2026;/usr
1780519292.9355682080;Wed Jun 3 22:41:32 2026;/usr/bin
1764774481.0000000000;Wed Dec 3 17:08:01 2025;/usr/bin/x86_64-linux-gnu-gold
1776448468.0000000000;Fri Apr 17 19:54:28 2026;/usr/bin/ntfsusermap
1648028710.0000000000;Wed Mar 23 11:45:10 2022;/usr/bin/bzless
1648202040.0000000000;Fri Mar 25 11:54:00 2022;/usr/bin/xdg-user-dirs-update
1769165477.0000000000;Fri Jan 23 12:51:17 2026;/usr/bin/mkdir
1707224063.0000000000;Tue Feb 6 14:54:23 2024;/usr/bin/expiry
1764774481.0000000000;Wed Dec 3 17:08:01 2025;/usr/bin/elfedit
```

Options:

- %T@ numeric timestamp
- %Tc human-readable time
- %p file path

Per-User Timeline

```
sudo find /home/kali/ -printf "%T@;%Tc;%p\n" > file_name  
sort -rn (file_name) | less
```

```
root@DESKTOP-6966M0H:/# sudo find /home/kali/ -printf "%T@;%Tc;%p\n" > timeline.csv  
find: '/home/kali/': No such file or directory  
root@DESKTOP-6966M0H:/# sudo find /home/kali/ -printf "%T@;%Tc;%p\n" > *.csv  
find: '/home/kali/': No such file or directory  
root@DESKTOP-6966M0H:/#
```

No such file or directory -r flag shows most recent changes first.

Expected Outputs of all command executed above

- **System info:** OS, kernel, uptime, user
- **SUID files:** List of privileged executables
- **Lynis audit:** Security warnings and compliance checks
- **Integrity checks:** Modified or corrupted files
- **Timeline:** Sorted file changes for forensic analysis